

Neural Topic Modeling

LLM Based selective extension and Mixture of Experts Routing

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Abstract

Neural topic models (NTMs) based on variational autoencoders (VAEs) often struggle to represent the full diversity of topics present in a corpus. a single encoder bottleneck routes representational capacity to majority classes and leaves rare labels no dedicated topic, semantically overlapping sub-categories drift onto the same top words and become indistinguishable, and the model cannot say which words drove a document to a given topic. This report summarises the progress of an investigation that addresses these failures from three complementary angles. **Thread 1** treats it as a data problem and uses LLMs to generate synthetic documents either for the rare classes alone or for every class, then measures the structural effect on the learned topic distribution. **Thread 2** treats it as an architectural problem and replaces the single VAE encoder with a Mixture-of-Experts (MoE) encoder governed by three document-level gating mechanisms (dense, sparse top- k , expert-choice). **Thread 3** pushes the architectural intervention deeper, introducing Parallel Word-Level Attention Experts (PWAE-NTM), a word-level attention-expert model where each expert attends over a document’s words with a learnable query and the gate routes documents based on those attended contexts.

1 Introduction

Topic modelling is an unsupervised technique that aims to discover latent topic structures within a corpus. It provides a dual output: coherent groups of words that represent topics, and distributions of these topics across documents. Neural Topic Models (NTMs) [2] cast topic discovery within an encoder-decoder framework trained end-to-end: the encoder maps each document to a topic distribution, and the decoder reconstructs the document through a learned topic-word structure. This design makes inference amortised and allows pretrained embeddings, attention, and auxiliary objectives to be incorporated as modular components. Some observed challenges faced by NTMs are:

- **Topic absence.** In imbalance datasets minority(rare) classes receive no dedicated topic as their vocabulary is either absorbed by a larger neighbouring class or scattered across multiple topics with no anchor. The resulting topic-word distribution shows no words that distinguishes rare-class documents from other classes.
- **Topic collision.** Two semantically distinct fields fuse into a single topic, so documents from

both classes produce numerically near-identical document-topic vectors and become inseparable.

- **Lack of interpretability.** Standard NTMs output a document-topic vector without explaining why a document received a particular topic weight. The encoder is a black-box MLP over a bag-of-words vector, and there is no per-word attribution showing which tokens drove the assignment.

This report covers three threads of work attacking these failures:

Thread 1 (data-augmentation angle, §3) addresses topic absence. The rare classes are dominated by other classes and lacks the signal that it needs to claim its own topic. The topic will be absent in the final output. Using an instruction-tuned LLM to extend the text of documents, either only for the rare classes (targeted) or for every class (blanket), and measure what the extension changes inside the learned topics and the visibility of the rare class topics. Importantly, the **rare topics** mentioned in this report refer strictly to the topics generated by rare classes, which is distinct from standard topics that merely receive a low probability in the document-topic distribution.

Thread 2 (architectural, §4) addresses topic absence and topic collision. MoE-NTM replaces the

single VAE encoder with a Mixture-of-Experts encoder, partitioning the latent space across experts so that rare-class and collided-class documents can be routed to dedicated capacity; then compare three document-level gating mechanisms (dense, sparse top- k , expert-choice).

Thread 3 (deeper architectural, §5) addresses lack of interpretability and gives a finer-grained handle on topic collision. PWAE-NTM pushes the routing inside the document where each expert attends over a document’s words via a learnable query and the gate routes based on attended word contexts. The assignments to a topic can be traced back to the specific words that drove them.

2 Background and Related Work

VAE-GSM [2] encodes a document into a Gaussian latent variable and applies a softmax topic layer and the decoder reconstructs the bag-of-words. ECRTM [3] adds embedding-clustering regularisation (unbalanced optimal transport [9] between topic vectors and word embeddings) to enforce topic diversity. FASTopic [4] replaces reconstruction with dual semantic-relation distillation between document, topic, and word embeddings. GloCOM [5] augments local document context with global aggregated context, particularly effective on short text.

Mixture-of-Experts in NTMs. MoE routing has been studied extensively for transformers (Switch [6], GShard) but remains under-explored for neural topic models. In a standard NTM the encoder compresses an entire document into a single latent vector; the resulting topic assignment offers no explanation of which words drove the document toward a particular topic. Introducing multiple experts, each attending over the document’s vocabulary with a learnable query, makes the routing decision inspectable: the per-expert attention weights reveal which words each expert considers salient, and the gating weights show how the experts are mixed for a given document. The goal is therefore interpretable topic assignment: not just that a document has certain topic proportions, but that a particular expert attended to a specific set of words and contributed a known weight to the final mixture.

LLM-based augmentation. The problem of data-sparsity in short text where there is too few word co-occurrences per document for a topic model to form topics. The rare classes are in even more problem-

atic position. PVTM [10] is the closest prior work: it uses an LLM to extend short texts into richer sequences and then trains a prefix-tuned language model coupled with a VAE for short-text topic modelling. This method creates richer signal for the topics, but because every document is extended uniformly, rare classes gain no relative advantage and continue to suffer. Here the same idea of using an LLM to supply the missing co-occurrence signal is studied as a controlled intervention: extended documents are generated either for the rare classes only (**+sel**) or for every class (**+all**), and the resulting changes inside the learned topics are examined.

Evaluation via downstream classification. Coherence metrics (NPMI, C_V) measure topic-word quality but cannot tell whether the learned document-topic vectors actually separate the semantic categories in the data. Manual topic-label alignment is impractical because topic indices are arbitrary: one model may place its energy topic at T_3 , another at T_8 , and a third may split it across two indices. A downstream classifier sidesteps this and it takes the full document-topic vector as input and learns whichever dimensions carry signal for each class, regardless of index ordering. This allows direct comparison across models without assuming any topic-to-label mapping.

3 Thread 1: LLM-Based Document Extension for Rare Classes

The rare classes are dominated by other classes and lacks the signal that it needs to claim its own topic. The topic will be absent in the final output. This thread asks whether synthetic documents from an instruction-tuned LLM can supply the missing mass, and whether they should be generated only for the rare classes (targeted) or for every class (blanket). It measures what the extension of documents changes inside the learned topics and the visibility of the rare class topics.

3.1 Problem Setting and Notation

The problem setting of topic modeling is recalled below, following LDA [1]. Consider a document collection X with V unique words (vocabulary size), each document denoted x . It wish to discover K latent topics from this collection. The k -th topic is defined as a distribution over all words (the topic-word distribution), denoted $\beta_k \in \mathbb{R}^V$; stacking these gives

$\beta = (\beta_1, \dots, \beta_K) \in \mathbb{R}^{V \times K}$, the topic-word matrix of all topics. The topic distribution of a document (the doc-topic distribution) records which topics it contains, denoted $\theta \in \Delta_K$, where Δ_K is the probability simplex $\Delta_K = \{\theta \in \mathbb{R}^K \mid \theta_k \geq 0, \sum_{k=1}^K \theta_k = 1\}$. The $d \in \mathbb{N}^V$ is used for the bag-of-words count vector of a document x (term frequencies tf_v , non-zero words $\{v_m\}$) and θ_d for that document’s doc-topic distribution.

3.2 Setup

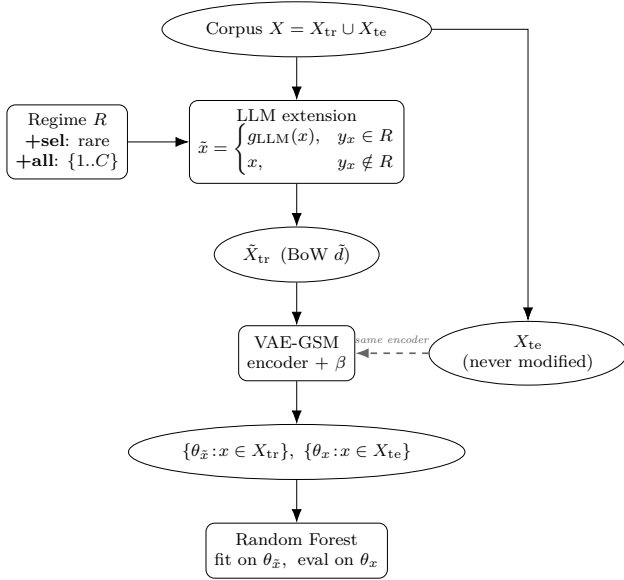


Figure 1: Text extension using LLM pipeline

Extension procedure. Using the notation of §3.1, let $X = \{x_1, \dots, x_N\}$ be the corpus with a fixed split $X = X_{\text{tr}} \cup X_{\text{te}}$ into a training set X_{tr} and a disjoint held-out test set X_{te} ($X_{\text{tr}} \cap X_{\text{te}} = \emptyset$), so the union $X_{\text{tr}} \cup X_{\text{te}} = X$ recovers the full corpus ($|X_{\text{tr}}| + |X_{\text{te}}| = N$), and let $y_x \in \{1, \dots, C\}$ be the class of document x . A regime fixes a *targeted class set* $R \subseteq \{1, \dots, C\}$: under **+sel**, R is the set of rare classes; under **+all**, $R = \{1, \dots, C\}$. Let g_{LLM} denote the per-document expansion that rewrites a short text into a detailed paragraph with LLM [11] (template in Appendix A). Each training document is mapped to

$$\tilde{x} = \begin{cases} g_{\text{LLM}}(x), & y_x \in R \text{ (targeted)}, \\ x, & y_x \notin R \text{ (kept as-is)}, \end{cases} \quad y_{\tilde{x}} = y_x,$$

so the label is preserved and $|\tilde{X}_{\text{tr}}| = |X_{\text{tr}}|$: expansion *replaces* a document’s text, it does not add documents. For example, on GoogleNews-10 under **+sel**,

$R = \{\text{climate_change, google_map}\}$, so a document x with $y_x = \text{climate_change} \in R$ is replaced by its expansion $g_{\text{LLM}}(x)$, whereas a China-class document ($y_x \notin R$) is left untouched. VAE-GSM is trained on the bag-of-words $\tilde{d}$ of the resulting corpus $\tilde{X}_{\text{tr}}$, yielding β and the doc-topic vectors $\theta_{\tilde{x}}$. The test set X_{te} is never modified; its documents are encoded by the *same* trained VAE-GSM to give their θ . The Random Forest is fit on $\{\theta_{\tilde{x}} : x \in X_{\text{tr}}\}$ and evaluated on $\{\theta_x : x \in X_{\text{te}}\}$ (Figure 1).

3.3 Datasets and Evaluation Protocol

Datasets. Four corpora are used across the three threads, summarised in Table 1. GoogleNews-10 and Tweet-10 are short-text benchmarks (6–9 words/document) used in Thread 1 (LLM extension), because their limited per-document vocabulary makes rare-class signal starvation most severe. Reuters-10 is the multi-label benchmark for Threads 2 and 3 (MoE-NTM and PWA-NTM); 20news_5class (≈ 188 words/document) serves as a second corpus for PWA-NTM in Thread 3.

Table 1: Datasets

Dataset	Docs	Avg. Words	Classes	Vocab
GoogleNews-10	2,414	6.3	10	500
Tweet-10	1,205	8.7	10	500
Reuters-10	2,929	187.8	10	2000
20news_5class	4,778	188.0	5	2000

Evaluation. Document-topic vectors $\theta \in \mathbb{R}^K$ are extracted from each trained model and used as features for Random Forest classification. The three metrics throughout: **RF macro-F1** ($\uparrow$): macro-averaged F1 from a Random Forest trained on the document-topic vector θ , measuring how well the topics separate the ground-truth classes. **NPMI** ($\uparrow$): normalised pointwise mutual information computed at the document level over a topic’s top-10 words against a Wikipedia reference corpus. Rewards globally distinctive word pairs. C_V ($\uparrow$): a sliding-window (110 tokens) coherence score that aggregates pairwise NPMI into per-word context vectors and takes their mean cosine similarity, capturing indirect word associations that document-level NPMI can miss. Correlates more strongly with human coherence judgements.

3.4 Quantitative Results

To see whether either regime recovers rare-class F1 and whether targeted extension beats blanket the eval-

uation is done on models as shown on the two short text corpora (Tables 2 and 3).

Table 2: GoogleNews-10.

Model	RF $\uparrow$	NPMI $\uparrow$	$C_V \uparrow$
LDA	0.747	0.194	0.434
ECRTM	0.791	0.187	0.412
FASTopic	0.984	0.193	0.409
GloCOM	0.994	0.171	0.418
PVTM	0.923	0.213	0.420
VAE-GSM-BoW	0.883	0.203	0.434
VAE-GSM-SBERT	0.776	0.208	0.422
VAE-GSM-SBERT+sel	0.950	0.211	0.437
VAE-GSM-SBERT+sel	0.989	0.206	0.422
VAE-GSM-BoW+all	0.977	0.242	0.436
VAE-GSM-SBERT+all	0.985	0.246	0.437

SBERT is variant where BoW as input is replaced by SBERT sentence embedding.

Table 3: Tweet-10

Model	RF $\uparrow$	NPMI $\uparrow$	$C_V \uparrow$
LDA	0.561	0.173	0.415
ECRTM	0.926	0.188	0.424
FASTopic	0.992	0.203	0.417
GloCOM	0.980	0.160	0.417
PVTM	0.839	0.227	0.408
VAE-GSM-BoW	0.962	0.191	0.424
VAE-GSM-SBERT	0.857	0.191	0.435
VAE-GSM-BoW+sel	0.917	0.198	0.423
VAE-GSM-SBERT+sel	0.994	0.199	0.411
VAE-GSM-BoW+all	0.922	0.214	0.418
VAE-GSM-SBERT+all	0.842	0.223	0.390

SBERT is variant where BoW as input is replaced by SBERT sentence embedding.

Targeted extension produces the best classifier on both datasets because it raises the per-document reconstruction cost of leaving a rare class without a dedicated topic, until allocation becomes the lower-loss option.

VAE-GSM distributes its K topics to minimise reconstruction loss over the training corpus (the reconstruction term $\mathcal{L}_{\text{recon}}$ of the ELBO). A rare class with N_{rare} training documents (the number of documents carrying that class label) contributes little: few documents and few words per document mean a generic neighbouring topic can absorb them at low total cost, and the model is free to spend the dedicated topic slot elsewhere. Extension under **+sel** keeps N_{rare} fixed but expands each rare-class document from a handful

of words to a detailed paragraph with class-distinctive vocabulary; the neighbouring topic now mis-predicts many more tokens per rare-class document, and the trade-off flips in favour of allocating a dedicated topic. The **+all** also expands the majority classes, but they already had enough mass to claim topics and the extra signal is redundant for them and the extension instead injects LLM stylistic generic vocabulary, consuming a topic slot for no class-separability gain.

The numbers track the mechanism. On Tweet-10, VAE-GSM-SBERT+sel (0.994) is the best classifier in the entire 11-model comparison, above FASTopic (0.992) and GloCOM (0.980). On GoogleNews-10, SBERT+sel closes 97% of the baseline-to-ceiling gap (0.776 $\rightarrow$ 0.989 vs. GloCOM 0.994). The highest-NPMI variants are the **+all** runs, not the best classifiers, which is the same point read from the other side: extending classes that already claim a topic improves coherence (tokens generated by the LLM do co-occur reliably) without improving separability. Coherence and classification are decoupled.

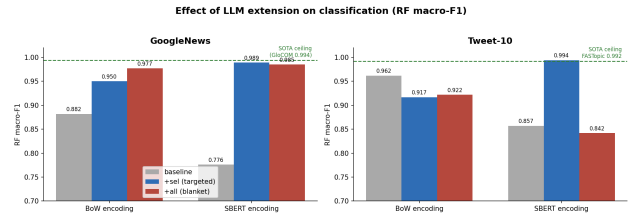


Figure 2: RF macro-F1 by encoding and extension regime

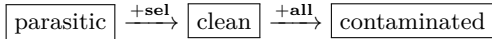
The **input encoding choice** interacts with label semantics: BoW input works well when label boundaries are defined by distinctive vocabulary (e.g. “wheat”, “crude”), while SBERT input works well when the distinguishing signal is contextual rather than lexical (e.g. climate_change vs. general China-policy news, which share many of the same words but differ in meaning). Against that backdrop one pattern is consistent across both datasets (Figure 2): every **+sel** case lifts its targeted rare class to F1 = 1.0 without dropping any targeted class below baseline, and **SBERT+sel** does so most reliably. **+all**, by contrast, is unreliable as on GoogleNews the rare classes hold near 1.0, but on Tweet-10 label_71 collapses from 1.000 to 0.500 under both encoders and label_20 drops to 0.333 under SBERT. The mechanism behind +sel’s advantage is structural rather than statistical: targeted extension injects the missing co-occurrence signal for rare classes without polluting the broader topic distribution, whereas blanket extension intro-

duces LLM-stylistic vocabulary that consumes topic capacity. This is analysed in detail in §3.5.

3.5 Qualitative Analysis: Top Topic words (β) Taxonomy

The top words from topic-word distribution lists give a way to interpret the topics based on the words. However on closer inspection, the words exposes pattern and noise in the words making it difficult to interpret the topics. Every observed regime change maps to one of five structural topic types, summarised in with examples from the trained models in Table 4. These words directly affects the NPMI and C_V scores. Analysing further gives the reason why the F1 numbers move and classification is affected.

Trajectory of topics top words: In GoogleNews-10 dataset the same rare classes like climate news and maps news moves through three taxonomy types as the approach changes:



Baseline- A single parasitic topic’s top words (row 2 in Table 4) hosts three classes at once (**China**, **climate_change**, **google_map**). The rare classes have no dedicated topic and are misclassified as the majority.

+sel- The parasitic topic splits into two clean topics, one per rescued class: $\{\text{google, view, street, airport, train, ...}\}$ for **google_map** and $\{\text{climate, change, refugee, new, zealand, ...}\}$ for **climate_change**. Each class now has its own topic.

+all- The two clean topics persist but acquire LLM-generated tokens (**white, paper; users, experience, technology, feature**). The class signal survives but the topic-word lists are contaminated by the LLM’s generated words.

The per-class F1 trajectory is the downstream consequence: **climate_change**: $0.20 \rightarrow 1.00 \rightarrow 0.88$; **google_map**: $0.80 \rightarrow 1.00 \rightarrow 1.00$. Both classes are rescued by +sel; **climate_change** then loses ground under +all as the contaminated tokens dilute the class signal.

Summary across all four VAE-GSM runs. Even though +all can match or exceed +sel on aggregate F1, the topics it produces are degraded: every

+all run contains at least one boilerplate or contaminated topic, while *no* +sel run does. On top of this, +all requires LLM generation for every class, so the compute cost scales with the number of classes rather than just the number of rare ones. +sel achieves the rare-class rescue at a fraction of the generation cost and without polluting the topic-word distribution.

3.6 Where the Extension harms

LLM extension (as mentioned in §3.2) does not uniformly help. Both regimes (+sel and +all) produce one failure each on Tweet-10, where the corpus is short enough that adding extended documents disturb the topic structure beyond the targeted rare-class rescue. Two cases:

- **Tweet-SBERT+all variant (0.842) drops below its baseline (0.857).** Blanket extension produces a boilerplate topic, and two rare classes (**label_71**, **label_79**) collapse into the same topic. Their θ vectors become near-identical and the classifier scores $F1 \approx 0.5$ on each. The aggregate loss from this collision exceeds the gains elsewhere.
- **Tweet-BoW+sel variant (0.917) drops below its baseline (0.962)** even though both *targeted* rare classes reach $F1 = 1.0$. Rescuing them displaces two *other* classes (**label_79**, **label_104**) into a shared topic, and the F1 loss on those two displaced classes outweighs the gain on the rescued ones.

Both failures share the same mechanism: extension reshuffles topic allocations, and on a corpus where every topic slot is already working hard, fixing one absent topic can collide two others into a single slot. Two recommendations follow. First, +sel remains the safer regime when the baseline shows rare-class topic absence (§3.4), since its failures are confined to side-effects of the rescue rather than broad topic degradation. Second, +all should be used with the quality of its top topic words in mind: even when its aggregate F1 is competitive, every +all run produces at least one boilerplate or contaminated topic (§3.5).

Table 4: Topic top words taxonomy

Type	Meaning	Example top words	Class assignment
Clean	One semantic field claims one topic; the topic’s documents come from a single class.	<i>{climate, change, refugee, new, zealand, status, decision, refugees}</i>	<code>climate_change</code> (5, 0 FN). Single class, no leakage.
Parasitic	A majority class dominates the topic and absorbs documents from one or more rare classes; rare-class documents are misclassified as the majority.	<i>{china, zone, air, japan, defense, disputed, google, climate}</i>	<code>China</code> (5) + <code>climate_change</code> (1) + <code>google_map</code> (5). Majority class swallows rare-class signal.
Collision	Two or more semantically distinct classes fuse into a single topic; their θ vectors become numerically near-identical and inseparable downstream.	<i>{king, speech, yemen, award, os-car, nomination, president, sag}</i>	<code>label_71</code> (4) + <code>label_79</code> (5) + <code>label_88</code> (5). Three classes fused.
Contaminated	A class-anchored topic acquires LLM-style vocabulary not present in the original corpus; the class signal survives but topic-word interpretability degrades.	<i>{google, view, street, airport, users, experience, technology, feature}</i>	<code>google_map</code> (5). Class signal preserved; bold tokens are LLM-introduced.
Boilerplate	A topic has no class anchor; it captures the LLM’s writing register (generic discourse vocabulary) rather than corpus content.	<i>{life, brown, home, just, text, heart, title, drive}</i>	No class loadings. Topic models LLM prose, not document content.

Notation: *class*(*n*) denotes *n* documents from that class assigned to that topic; FN = false negatives.

3.7 Coherence Does Not Predict Classification

The **+all** variants post the highest NPMI in both datasets, yet neither is the best classifier the two metrics point in different directions. A higher NPMI carries almost no information about whether the topics separate the classes.

The reason is what NPMI rewards. NPMI scores how reliably a topic’s top words co-occur, not whether those words discriminate between classes. The LLM-style vocabulary that **+all** injects is internally coherent in exactly this sense: generic tokens like “title, conclusion, approach” co-occur reliably across the LLM’s generated text, so they inflate NPMI while carrying no class signal. **+all** therefore buys coherence without buying class separability, and **NPMI must not be used to select the extension regime** instead the regime choice should be made on RF macro-F1 (separability) and on the top-word taxonomy (§3.5, interpretability).

3.8 Hinge-Gated β -Diversity: A Local Fix for Post-Extension Collapse

LLM extension successfully rescues the targeted rare classes (§3.4) but sometimes leaves a residual collapse between two unrelated topics whose top word is shared. This subsection introduces a local modification to VAE-GSM’s diversity regulariser that removes the collapse.

Observation. VAE-GSM’s existing diversity regulariser saturates while topic collapse persists in vocabulary space. Following Miao et al. [2] (Appendix B), the regulariser operates on the latent topic vectors $v_k \in \mathbb{R}^D$ ($D = 256$), not on the topic-word rows β_k . It measures the pairwise angle between topic vectors,

$$a(v_i, v_j) = \arccos\left(\frac{|v_i \cdot v_j|}{\|v_i\| \|v_j\|}\right),$$

takes the mean ζ and variance ν over all distinct topic

pairs,

$$\zeta = \frac{1}{K(K-1)} \sum_{i \neq j} a(v_i, v_j),$$

$$\nu = \frac{1}{K(K-1)} \sum_{i \neq j} (a(v_i, v_j) - \zeta)^2,$$

and *subtracts* $\lambda(\zeta - \nu)$ from the training objective ($\lambda = 0.1$), equivalently minimising

$$L_{\text{div}} = -(\zeta - \nu),$$

which pushes the mean inter-topic angle up and its variance down so the topic vectors spread out in $\mathbb{R}^D$. With only $K = 10$ vectors in $D = 256$ dimensions, mutual orthogonality is cheap: L_{div} saturates at its floor of $-\pi/2 \approx -1.57$ early in training and stays flat for the rest of the run.¹ The loss reports success while topic collapse persists in *vocabulary* space

Change. Move the penalty into vocabulary space that is directly onto the topic-word rows $\beta_k \in \mathbb{R}^V$ and gate it with a hinge so it fires only on collapsed pairs. For topics $i \neq j$ let

$$s_{ij} = \cos(\beta_i, \beta_j) = \frac{\beta_i^\top \beta_j}{\|\beta_i\|_2 \|\beta_j\|_2} \in [0, 1]$$

4 Thread 2: MoE-NTM with Document-Level Gating

Thread 1 rescued rare classes by adding data; this thread asks whether the same rescue can come from *architecture* instead, and targets both *topic absence* and *topic collision*. A standard NTM compresses each document through a single VAE encoder, so representational capacity flows to the majority classes: a rare class gets no dedicated latent direction (absence), and two distinct fields can fuse into one (collision). MoE-NTM replaces that single encoder with a Mixture-of-Experts encoder, partitioning the latent space across E experts so that rare-class and collided-class documents can be routed to dedicated capacity. The investigation is whether MoE actually buys that capacity, and which *document-level* gating rule (dense, sparse top- k , or expert-choice) delivers it.

4.1 Architecture

We use the notation of §3.1: $d \in \mathbb{N}^V$ is the bag-of-words count vector of a document over a vocabulary

be their cosine similarity. $s_{ij} \in [0, 1]$ because the topic-word rows are non-negative ($\beta_{k,v} \geq 0$, $\sum_v \beta_{k,v} = 1$); $s_{ij} = 1$ iff $\beta_i = \beta_j$. The regulariser penalises only the *excess* similarity above a threshold τ ,

$$L_{\beta\text{-hinge}} = \frac{2}{K(K-1)} \sum_{i < j} (s_{ij} - \tau)_+,$$

where $(x)_+ = \max(x, 0)$, and enters the objective as the added term $\lambda L_{\beta\text{-hinge}}$. The hinge $(\cdot)_+$ zeroes the gradient on every pair with $s_{ij} \leq \tau$.

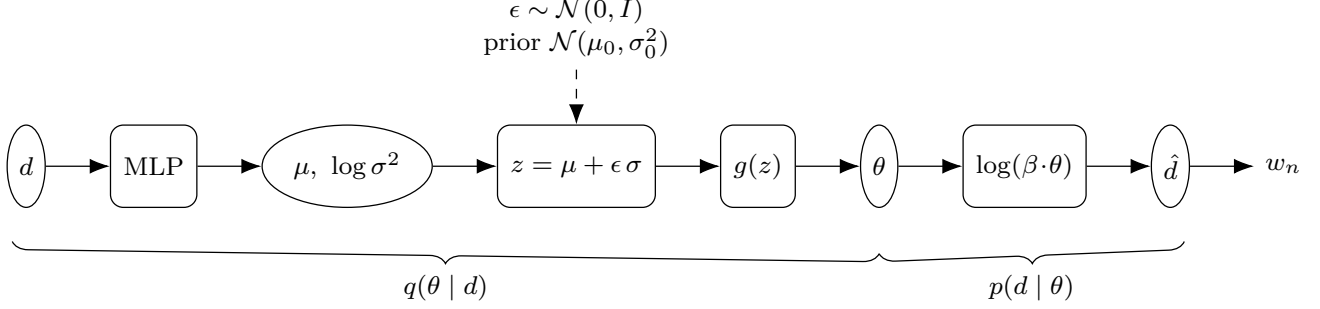
Results: On `tweet_10+sel` (seed 42) the hinge recovers the collapsed *King's Speech*/diet pair: NMI improves $0.948 \rightarrow 0.983$, Purity $0.903 \rightarrow 0.991$, and RF macro-F1 $0.994 \rightarrow 0.995$, with the Oscars tokens (*king, speech, oscar, award, nomination*) purged from the diet topic (Appendix B). On `googlenews+sel` (no collapse) RF moves only ± 0.002 . Two failure modes remain: the hinge cannot separate genuine semantic subtopics that legitimately share a dominant word (reconstruction re-anchors it), and on large vocabularies ($V = 2000$) reconstruction dominates the hinge term and collapse can persist. It is a fix for small-vocabulary post-extension collapse, not a general diversity regulariser.

of size V , $\theta \in \Delta_K$ is its doc-topic distribution over K topics, and $\beta \in \mathbb{R}^{V \times K}$ is the topic-word matrix.

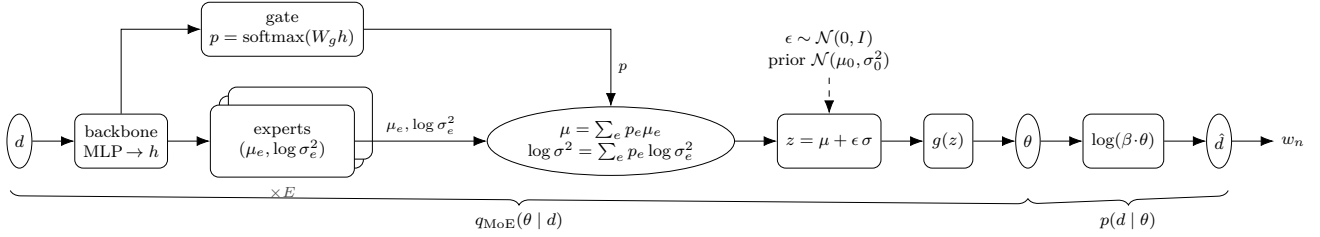
Baseline (VAE-GSM). The baseline VAE-GSM [2] (Figure 3(a)) encodes each document with a *single* amortised inference network. An MLP maps the (L1-normalised) input d to the parameters $(\mu, \log \sigma^2)$ of a diagonal Gaussian; a latent $z = \mu + \epsilon \odot \sigma$ with $\epsilon \sim \mathcal{N}(0, I)$ is drawn by the reparameterisation trick; the doc-topic distribution $\theta = \text{softmax}(g(z)) \in \Delta_K$ is obtained from a linear topic layer g ; and the bag of words is reconstructed through β as $\hat{d} = \text{softmax}(\beta\theta)$. Training maximises the ELBO of $\log p(d)$, factorised as the encoder $q(\theta | d)$ and the decoder $p(d | \theta)$. Because the entire encoder is one bottleneck, representational capacity is shared across all classes that is the structural cause of the rare-class *absence* and *collision* failure.

MoE encoder. MoE-NTM leaves the decoder and the $z \rightarrow \theta \rightarrow \beta$ path untouched and replaces only the single encoder with a mixture-of-experts encoder (Fig-

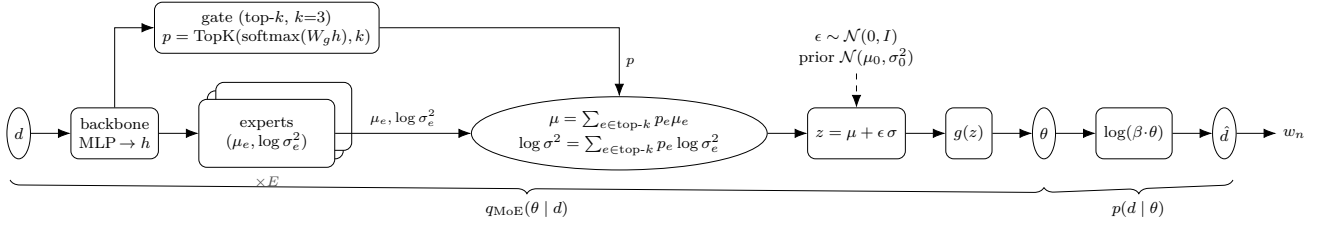
¹The floor is reached when all topic vectors are mutually orthogonal, where $\arccos(0) = \pi/2$ and the loss has no gradient left.



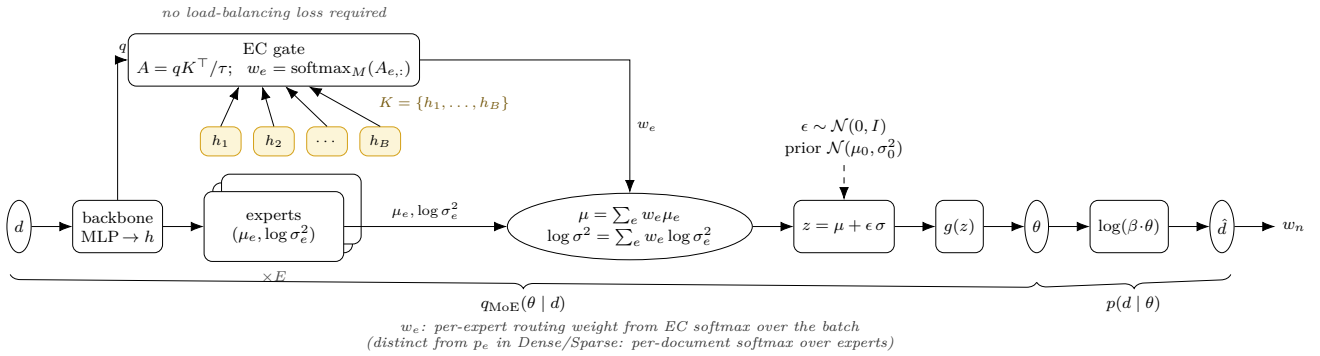
(a) Neural variational inference (VAE-GSM baseline).



(b) Dense MoE-NTM (all E experts active; softmax gate).



(c) Sparse top- k MoE-NTM ($k = 3$; only k experts active).



(d) Expert-Choice MoE-NTM (experts select documents; routing inverted).

Figure 3: MoE-NTM architectural lineage (a) VAE-GSM baseline [2]. (b) Dense MoE-NTM. (c) Sparse top- k . (d) Expert-Choice [7].

ure 3(b)), following the standard MoE formulation [8]: (i) a shared backbone MLP maps d to a hidden vector h ; (ii) E expert sub-networks each map h to their own Gaussian parameters $(\mu_e, \log \sigma_e^2)$, $e = 1, \dots, E$; (iii) a gating network forms weights $p = (p_1, \dots, p_E)$ from h , and the per-expert posteriors are combined into a single Gaussian

$$\mu = \sum_e p_e \mu_e, \quad \log \sigma^2 = \sum_e p_e \log \sigma_e^2,$$

which re-enters the unchanged $z \rightarrow \theta \rightarrow \hat{d}$ path. Different experts can thus specialise on different regions of the label space while the decoder β stays shared. Only the *routing rule* that is how the weights p_e are formed distinguishes the three variants evaluated here:

- **Dense** (Figure 3(b)). $p = \text{softmax}(W_g h)$: all E experts are active for every document. A load-balancing loss is needed to stop the gate collapsing onto a few experts.
- **Sparse top- k** (Figure 3(c)). $p = \text{TopK}(\text{softmax}(W_g h), k)$: the gate keeps only its k largest entries (zeroing the rest), so only k of the E experts are active per document. A load-balancing loss is still required.
- **Expert-Choice (EC)** [7] (Figure 3(d)). EC scores *documents* per expert. Each expert e holds a learnable query q_e ; the gate scores it against the hidden states $K = \{h_1, \dots, h_B\}$ of all B documents in the batch via $A = qK^\top / \tau$, then softmaxes across the batch to produce routing weights $w_e = \text{softmax}_M(A_{e,:})$. Each expert keeps its top- M documents ($M = \lceil B \cdot \text{cf} / E \rceil$, where cf is the capacity factor; using cf = 1.5, gives each expert 50% overcapacity beyond the even split B/E so a document may be selected by several experts). Because per-expert capacity is fixed, every expert receives exactly M documents per batch and no load-balancing loss is required.

4.2 Results on Reuters-10

To determine which gating rule best separates the classes and where any gain originates comparison is done of the three mechanisms against the VAE-GSM baseline in (Table 5).

Table 5: Reuters-10 results . Best per column **bold**.

Model	RF	micro	NPMI	TopDiv
VAE-GSM (baseline)	0.691	0.796	0.222	0.72
MoE, dense	0.686	0.790	0.250	0.81
MoE, sparse top- k	0.690	0.797	0.256	0.80
MoE, expert-choice	0.724	0.802	0.250	0.77

Expert-Choice achieves the best RF macro-F1 (0.724) and micro-F1 (0.802). The gain over the next-best MoE variant (sparse, 0.690) comes almost entirely from one tail label:

Table 6: Tail-label F1: previous best vs. Expert-Choice.

Label	Supports	Prev best	EC	Δ
nat-gas	21	0.250	0.546	+0.296
bop	20	0.737	0.757	+0.020
oilseed	35	0.464	0.464	0.000
dlr	33	0.623	0.660	+0.037

Why EC works on rare label (nat-gas). Under token-choice routing, nat-gas documents (1–2 per batch) compete with crude documents for the same expert slots and no expert specialises on the minority. EC inverts the direction: each expert is forced to process exactly C documents per batch regardless of class frequency, so the minority gets guaranteed coverage. EC is the only model to clear the 0.25 F1 ceiling on nat-gas.

Expert-Choice fixes tail-class capacity, but its gate still routes on the opaque document vector h and cannot say *which words* sent a document to an expert. Thread 3 keeps the mixture-of-experts idea but moves routing to the word level to recover that interpretability.

5 Thread 3: PWAE-NTM (Parallel Word-Level Attention Experts)

5.1 Motivation

Building directly on Thread 2, PWAE-NTM keeps the mixture-of-experts encoder but changes *where* routing happens. Thread 2 routes at the *document* level: a document is encoded into a single hidden vector h by a shared backbone and the gate operates on h . This collapses the whole document into one opaque vector

before routing, so it offers no way to interpret what drives a document to a particular expert or topic. The gate sees only h , not the words; we cannot recover which tokens were responsible for the assignment, even though the distinguishing signal for tail classes is typically carried by a small subset of the document’s words (e.g. “pipeline”, “barrel” for crude).

PWAE-NTM (Pairwise-Attention Word-Attended Expert NTM) makes the routing *word-level* precisely to close this interpretability gap: each expert e has a learnable query vector q_e that attends over the document’s non-zero words, producing per-word attention weights α_e and an attended context c_e ; the gate then routes on $\{c_1, \dots, c_E\}$. Because routing is built from explicit per-word attention, a document’s assignment to an expert and topic can be traced directly back to the *topic-deciding words* that caused it to be routed there, recovering the interpretability the document-level gate discards.

5.2 Architecture

Figure 4 shows the PWAE-NTM architecture. One expert pathway is drawn explicitly in the figure; the stacked box runs E times in parallel, one copy per expert.

All experts share a single *frozen* word-embedding matrix $\Phi \in \mathbb{R}^{V \times H}$, where V is the vocabulary size and row $\phi_v \in \mathbb{R}^H$ is the embedding of vocabulary word v . A document d is represented by its set of *non-zero word indices* $\{v_1, \dots, v_M\}$ (M = number of distinct words actually present in d), together with their per-document TF-IDF masses tfidf_m .

Each expert e owns one *learnable query vector* $q_e \in \mathbb{R}^H$. The query scores every word of the document by its dot product with that word’s embedding, divided by a temperature τ , and the scores are normalised over the document’s M words:

$$\alpha_{e,m} = \text{softmax}_m \left(\frac{q_e \cdot \phi_{v_m}}{\tau} \right), \quad \sum_m \alpha_{e,m} = 1.$$

The *attention weight* $\alpha_{e,m} \in [0, 1]$ says how strongly expert e attends to the m -th word of d ; a small τ sharpens the attention onto a few words. These weights are exactly what makes the routing interpretable (§5.3).

Expert e collapses the words it attends to into a single *context vector*, pooling their embeddings weighted by both the attention and the TF-IDF mass:

$$c_e = \sum_m \alpha_{e,m} \text{tfidf}_m \phi_{v_m} \in \mathbb{R}^H.$$

c_e is expert e ’s H -dimensional summary of the document seen through its own attention; the tfidf_m factor down-weights frequent discourse words so the context is anchored on content words.

The gate decides *how much each expert speaks* for this document. It concatenates the E contexts and produces one weight per expert through $W_g \in \mathbb{R}^{E \times (E \cdot H)}$:

$$g = \text{softmax}(W_g [c_1 \parallel \dots \parallel c_E]) \in \Delta^E.$$

The final document-level topic vector is the gated, L_1 -normalised mixture of the per-expert distributions. The variational encoder was replaced by design choice:

$$\theta_d = \frac{\sum_e g_e \theta_e}{\|\sum_e g_e \theta_e\|_1} \in \Delta^T.$$

Training and Regularisation:

The desing choice to remove variational encoder (the KL term was removed after a posterior-collapse failure in an earlier variant);

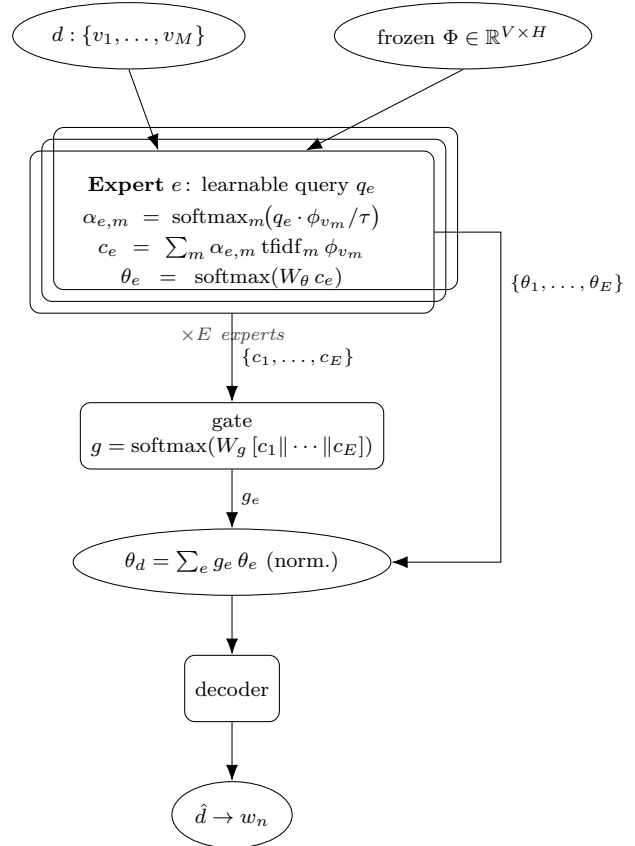


Figure 4: PWAE-NTM architecture (one of E experts shown explicitly)

queries q_e are k -means-initialised on TF-weighted document embeddings; four regularisers (attention-diversity, θ -diversity, topic-diversity, and a negative gate-entropy term $\lambda_{\text{gate}} = -0.05$ that encourages each document to commit to few experts) plus batch-level load balancing keep the experts from collapsing onto one another.

5.3 Word-Level Interpretability: Worked Examples

The central motivation of §5, that word-level attention makes routing *interpretable*, is best shown on single documents. We trace one document from each dataset end-to-end.

20news-5. Figure 5 traces the `rec.motorcycles` document “*i wanted to let people know that this motorcycle has been sold*” (20news-5class, $E = T = 5$). The gate concentrates all mass on a single expert ($g_{E0} = 1.00$, vs. uniform $1/E = 0.20$); its attention, weighted by $\alpha \times \text{idf}$, lands almost entirely on *motorcycle* (0.84), the obvious class cue. $E0$ feeds the motorcycle topic $T4$ (*bike, ride, riding, motorcycle, bmw*), so the chain *document* $\rightarrow$ *attended words* $\rightarrow$ *expert* $\rightarrow$ *topic* is read directly off the attention weights.

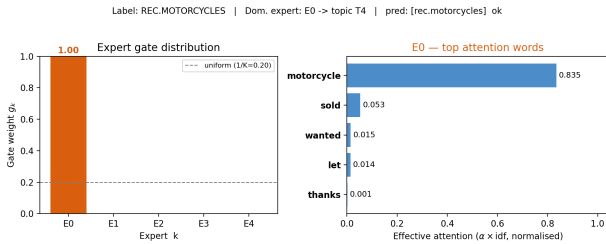


Figure 5: Word-level routing walkthrough for a `rec.motorcycles` document (20news-5). **Left:** the gate places all mass on $E0$. **Right:** $E0$ attends almost exclusively to *motorcycle*, the obvious class cue.

Reuters-10. The same end-to-end traceability holds on the denser financial corpus. Figure 6 traces an **interest**-class headline, “*Midland Bank said it cutting its base rate to 10.5 pct from 11 pct*” (doc 1774, correctly classified). The gate concentrates almost all mass on $E1$ ($g_{E1} = 0.88$ vs. uniform 0.10); $E1$ ’s attention lands on *midland* (0.72), *base* (0.17) and *bank* (0.09), the exact tokens a human would use to tag central-bank interest-rate news, and $E1$ feeds topic $T6$ (*bank, pct, billion, rate, mln, stg, fed*). A document-level gate (Thread 2) routes on the opaque hidden vector h and exposes none of this: there is no

way to recover that “midland” and “base rate” were the deciding tokens.

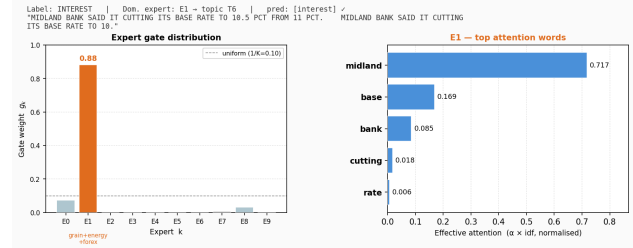


Figure 6: Word-level routing walkthrough for an **interest** document (doc 1774, Reuters-10). **Left:** the gate places 0.88 of its mass on expert $E1$. **Right:** $E1$ ’s top attended words by effective attention ($\alpha \times \text{idf}$) are *midland*, *base*, *bank*. These are exactly the interest-rate cues, making the routing decision human-readable.

5.4 Corpus-Level Routing Structure

The section §5.3 traced individual documents; here inspection is by routing over the *whole* corpus. For each document, record two things: its *dominant expert* – the expert with the largest gate weight and its *dominant topic* – the topic with the largest mass in the mixture. Tabulating how dominant experts map to dominant topics across all documents (the expert $\rightarrow$ topic map, Figures 7 and 8) exposes the global routing structure, and two patterns recur on both datasets: (i) the mapping is *near-deterministic* as almost every document handled by a given expert lands on the same topic; and (ii) a *hub topic* absorbs several experts at once.

20news-5. The structure (Figure 7) shows three effects. (i) Each expert maps to a near-deterministic topic ($E0 \rightarrow T4$ 96%, $E2 \rightarrow T3$ 99%, $E4 \rightarrow T1$ 97%). (ii) A *hub* topic emerges: $T0$ (sports, *game/team/hockey/baseball*) absorbs *both* $E1$ and $E3$ at 100%, yet the two experts attend to *different* word sets (Table 7): $E1$ fires on named entities (teams, players) while $E3$ fires on box-score vocabulary (*pts, season, period*). (iii) The garbage verb-cluster topic $T2$ (*finished, finish, sending, kept*) is an *orphan*. No expert routes to it as dominant.

Reuters-10. The same structure appears on the larger Reuters corpus, only more pronounced (Figure 8). The expert–topic link is again near-deterministic. $E1$ ’s dominant topic is $T6$ in **all 161** documents it dominates (100%), and the same holds for every expert ($E6 \rightarrow T3$ 99%, $E4 \rightarrow T8$ 97%, $E5 \rightarrow T5$ 97%). The hub is even stronger: $E0$, $E1$ and $E7$

all route to $T6$ at 100% and $E8$ at 60%, so the financial topic $T6$ (*bank/rate/fed*) is over-subscribed by four experts. They are not redundant. They attend to distinct sub-vocabularies as shown in Table 8, but collapse onto one coarse topic.

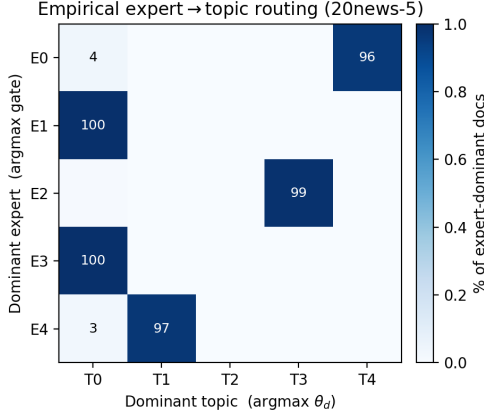


Figure 7: Empirical expert → topic routing on 20news-5.

Table 7: The two experts that collapse onto the sports hub $T0$ attend to *distinct* word sets: $E1$ on named entities, $E3$ on box-score vocabulary. The shared-document row shows both experts attending to the *same* sentence. $E1$ picks the team/player names, $E3$ picks the temporal/season frame, yet both feed $T0$. Words ranked by $\alpha \times \text{idf}$.

Shared document (routed to $T0$): “brad mccrimmon was the captain of the flames when he was traded to detroit following the 1989–90 season”

$E1$ attends *flames, brad, captain, detroit*
 $E3$ attends *season, following*

Top attended words ($\alpha \times \text{idf}$, corpus):

$E1$ (entities) baseball, braves, cubs, hockey, pitcher, flyers
 $E3$ (box-score) season, year, period, pts, april, points

$T0$ (β top-10): *game, team, games, year, season, hockey, baseball, player, teams, nhl*

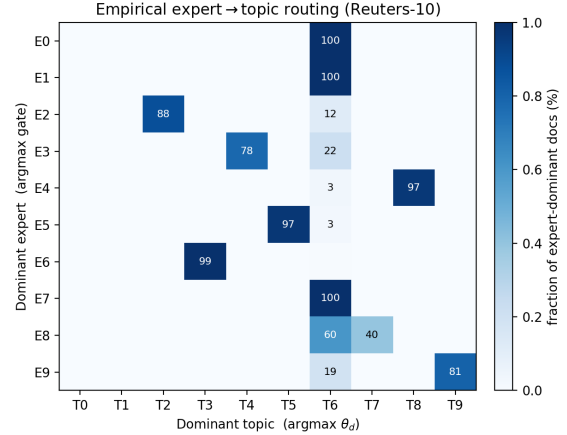


Figure 8: Empirical expert → topic routing on Reuters-10.

Table 8: The four experts that collapse onto topic $T6$ attend to *distinct* financial sub-vocabularies. Words ranked by effective attention ($\alpha \times \text{idf}$) over the documents each expert dominates; → $T6$ is the routing fraction from Figure 8.

Expert	→ $T6$	Top attended words	Labels
$E0$	100%	stg, assistance, drain, balances	money-fx
$E1$	100%	prime, fed, base, midland, bundesbank	interest
$E7$	100%	billion, rate, dollar, funds, treasury	interest
$E8$	60%	feb, february, april, january, march	dates
$T6$ (β top-10) shared topic		<i>bank, pct, billion, rate, mln, stg, fed, rates, dlrs, banks</i>	

5.5 Domain-Specialist Quantification

The worked examples of §5.3 generalise that when PWAE-NTM routes a document confidently, the chosen expert attends to that document’s *domain* vocabulary rather than to generic discourse words, and this holds across the corpus on both (Reuters-10 and 20news-5class) datasets.

The intuition is simple. A single worked example can always be cherry-picked, so a corpus-wide number that tests the same claim is needed: *when an expert decides a document belongs to it, is it looking at the words that actually define that topic, or at incidental discourse?* If an expert’s attention concentrates on the model’s own topic-defining words far more than

it would be by chance, the routing is grounded in topic semantics rather than surface noise, and the explanation that is read off the attention is trustworthy. Therefore a metric is needed to measure the overlap between *where an expert attends* and *what the decoder treats as topic content*, relative to what pure chance would give.

The effect is quantified against the model’s own **topic-defining vocabulary** $\mathcal{V}_{\text{topic}}$, the union of the top-50 decoder (β) words across all topics, i.e. the words that actually define what the topics are. An expert’s attention is **domain-relevant** if it lands on $\mathcal{V}_{\text{topic}}$. For each confidently-routed document d (dominant gate weight $g_{k^*} > 0.6$, where $k^* = \arg \max_k g_k$), the *observed* domain-relevant fraction is the share of the dominant expert’s effective attention ($\alpha \times \text{idf}$) that falls on $\mathcal{V}_{\text{topic}}$:

$$\text{obs}(d) = \frac{\sum_{m: v_m \in \mathcal{V}_{\text{topic}}} \alpha_{k^*,m} \text{idf}_{v_m}}{\sum_m \alpha_{k^*,m} \text{idf}_{v_m}}.$$

The corresponding *chance* fraction is what the same statistic would equal under uniform attention over d ’s non-zero words:

$$\text{chance}(d) = \frac{\sum_{m: v_m \in \mathcal{V}_{\text{topic}}} \text{idf}_{v_m}}{\sum_m \text{idf}_{v_m}}.$$

Both numerator and denominator carry the same idf weighting, so the lift compares like with like: how the gate’s idf-weighted attention distributes over d ’s words versus how a uniform idf-weighted attention would distribute over the same words. The **lift** is $\text{obs}(d)/\text{chance}(d)$, averaged per label over confidently-routed documents (Table 9); the document-specific $\text{chance}(d)$ controls for each document’s own word composition.

The gate (W_g) and decoder (β) are trained jointly but untied (no shared weights, routing loss never sees β), so the gate *could* route on surface features unrelated to topic meaning, giving lift $\approx 1\times$. However it does not as seen the $3.1\times$ (Reuters) and $2.8\times$ (20news) lifts in Table 9 show the gate attends to the same words the decoder uses to define each topic. Routing is thus not only readable (§5.3) but *faithful* – the document→words→expert→topic chain genuinely explains the decision.

Table 9 reports the lift for every label. Overall, confident routing is $3.1\times$ more domain-relevant than chance on Reuters and $2.8\times$ on 20news, and the weakest per-label lift is religion at $1.61\times$, still substantially above chance; the effect is not driven by a few categories. The lift does vary by label, tending to be highest where a label owns a distinctive content

lexicon (Reuters bop $3.97\times$, money-fx $3.57\times$; 20news hockey $3.86\times$, motorcycles $3.63\times$) and lowest for categories whose documents share more generic vocabulary (Reuters dlr $2.10\times$, nat-gas $2.46\times$; 20news religion $1.61\times$, crypt $1.92\times$).

Table 9: Per-label domain-relevant attention lift for confidently-routed documents (dominant gate > 0.6).

Label	n	Lift
bop	6	$3.97\times$
money-fx	164	$3.57\times$
interest	102	$3.43\times$
oilseed	43	$3.27\times$
trade	17	$3.22\times$
ship	152	$3.12\times$
grain	223	$3.11\times$
crude	234	$2.97\times$
nat-gas	41	$2.46\times$
dlr	35	$2.10\times$
Reuters-10 (overall)		$3.1\times$
hockey	632	$3.86\times$
motorcycles	675	$3.63\times$
baseball	462	$3.26\times$
crypt	676	$1.92\times$
religion	776	$1.61\times$
20news-5 (overall)		$2.8\times$

5.6 Limitations: Garbage Topics

Two topics in the best run, T0 and T1, are not semantic clusters but clusters of *corporate-action verbs* (*completed*, *expanded*, *revised*, *stabilise*), which TF-IDF attention amplifies because these tokens have moderate-to-high IDF (3.4–4.8). Two topic slots are spent on this verbal-register axis rather than on content, which costs the smallest classes (bop, nat-gas, oilseed) representational capacity.

6 Key Findings and Conclusion

This report investigated three failure modes of neural topic models under class imbalance – topic absence, topic collision, and lack of interpretability across three complementary threads. The principal findings are:

1. **A common topic-failure taxonomy.** The five-way taxonomy of Thread 1 (clean, parasitic, collision, contaminated, boilerplate) classifies failure modes across all three threads, including PWAE-NTM’s IDF-amplified garbage

topics. The same diagnostic lens applies regardless of the underlying cause.

2. **Topic collapse lives in vocabulary space.** Angular diversity on latent topic vectors saturates once they are orthogonal, yet two topics can still share a top word. A hinge-gated penalty in β -space resolves collapse where it actually occurs and remains inert when no collapse exists.
3. **Coherence and classification are decoupled.** NPMI does not predict downstream classification quality. The +all variants of Thread 1 post the highest NPMI but are not the best classifiers, and PWAE-NTM has the highest NPMI on Reuters-10 yet trails MoE-NTM-EC on macro-F1. Both metrics should be reported.
4. **Tail-class rescue can be achieved through different levers.** Thread 1 supplies training mass to rare classes via targeted LLM extension. Thread 2 guarantees per-expert capacity through Expert-Choice routing. Thread 3 drives experts to disjoint attention vocabularies at the word level.
5. **Word-level routing is interpretable and faithful.** PWAE-NTM’s per-word attention traces each topic assignment to specific tokens. Confidently-routed documents attend to the topic-defining vocabulary at $2.8\text{--}3.1\times$ above chance, and this lift demonstrates agreement between the gate and the decoder despite their being untied parameters.

The thread connecting all five findings is that topic models become more controllable when their internal structure is made visible that shows which words anchor a topic, which tokens route a document, and where in the model collapse has occurred. Making that structure visible is not only an interpretability goal it also opens a scope for targeted intervention.

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APPENDIX

A LLM Extension Prompt (Section 3)

Each short document is expanded by Mistral-7B-Instruct-v0.3 with the following single-turn instruction (the document text is substituted for `{text}`); generation uses a 500-token cap per document.

Given the short text: "`{text}`", expand it into a detailed paragraph that provides background and elaborates on the key points to enrich its context. Try to make it as detailed as possible.

B Word-Level Impact of Hinge-Gated β -Diversity (Section 3.8)

Table 10 shows the two *affected* topics on `tweet_10+sel` (seed 42) *before* and *after* applying the hinge-gated β -diversity loss ($\tau = 0.3$, $\lambda = 0.2$, activated at epoch 150). The remaining eight topics are unchanged (only minor tail-word reordering) and are omitted. Words in **bold** are the *King’s Speech* / Oscars tokens (*king*, *speech*, *oscar*, *award*, *nomination*) that, before the fix, leaked into the unrelated acai-berry diet topic — the two topics shared *king* as their top word (argmax collision, $\cos(\beta_i, \beta_j) \approx 0.9$).

The hinge fires only on this one collapsed pair. After the fix the diet topic sheds all Oscars vocabulary (*acai*, *berry*, *weight*, *loss*, *diet*, *plan*, *healthy*, *lose*, *reduction*, *juice*) and the Oscars topic consolidates (*king*, *speech*, *award*, *oscar*, *nomination*, *sag*, *academy*). NMI $0.948 \rightarrow 0.983$; Purity $0.903 \rightarrow 0.991$; RF macro-F1 $0.994 \rightarrow 0.995$.

Table 10: Top-10 words of the two collapsed topics before vs. after hinge-gated β -diversity (`tweet_10+sel`, seed 42). Shared Oscars tokens (bold) leak into the diet topic before the fix and are removed after.

Topic	Before (no hinge)	After (hinge)
Diet (acai)	acai, berry, king , speech , weight, loss, diet, award , oscar , nomination	acai, berry, weight, loss, diet, plan, healthy, lose, reduction, juice
Oscars	speech , king , award , oscar , fishing, sag, academy, nomination , commercial, best	king , speech , award , oscar , nomination , sag, academy, best, lead, win